Parametric Equations & Modeling

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.03

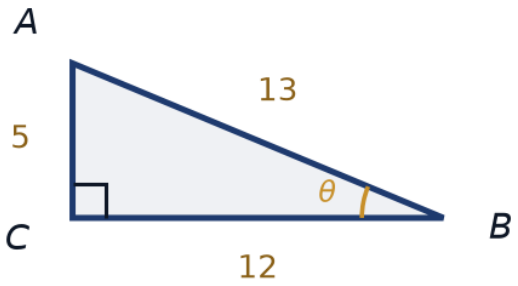
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating parametric equations

Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.

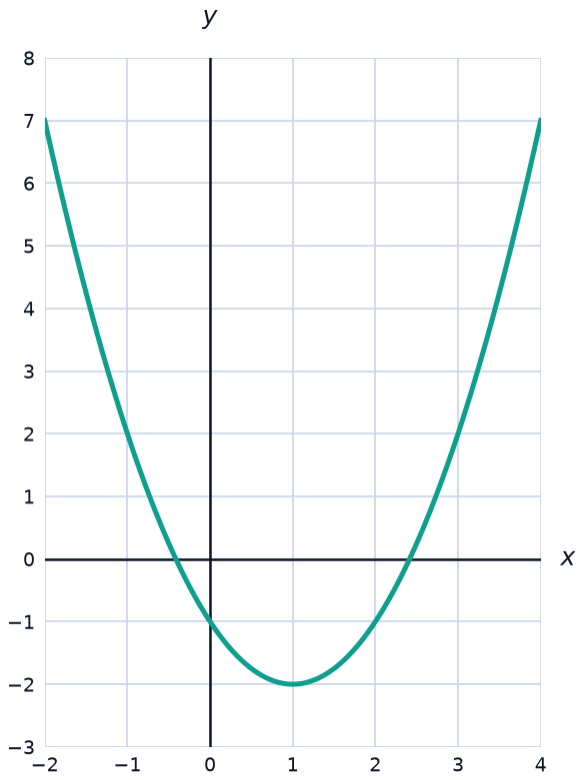
2 [Foundational] • Target: Evaluating parametric equations

Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.



3 [Foundational] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.



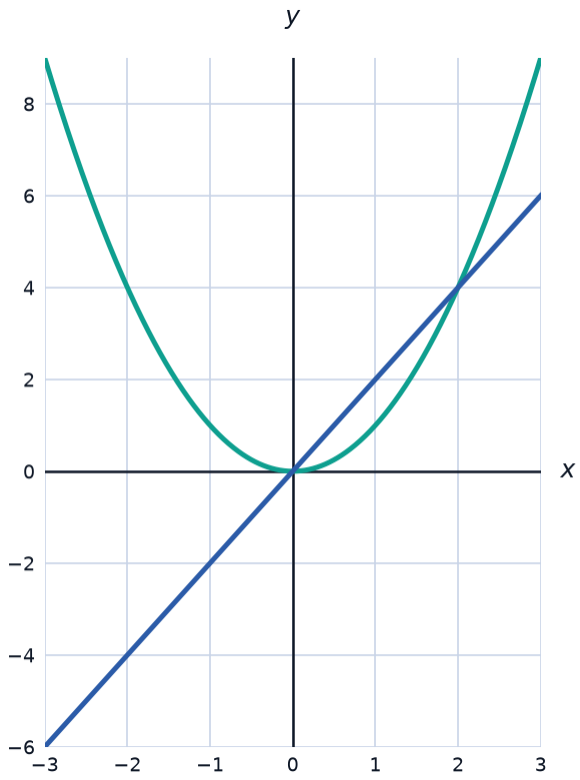
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4 [Foundational] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.

5 [Foundational] • Target: Visualizing a parametric curve's components

The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.



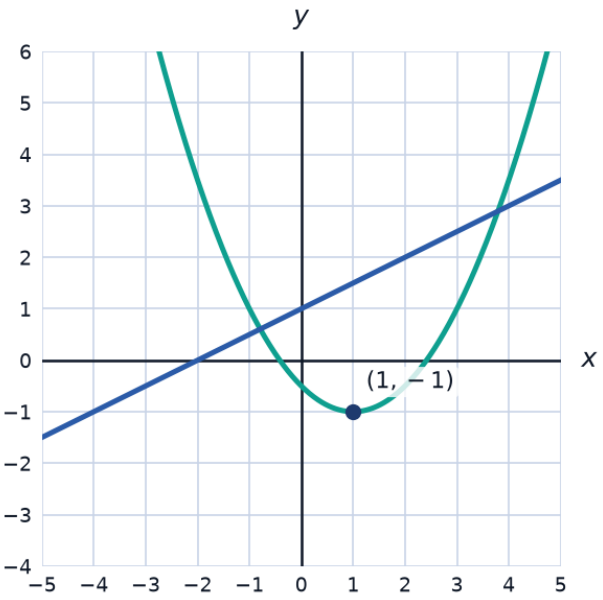
a Eliminate the parameter t to find the rectangular equation relating x and y for this curve.

b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown).

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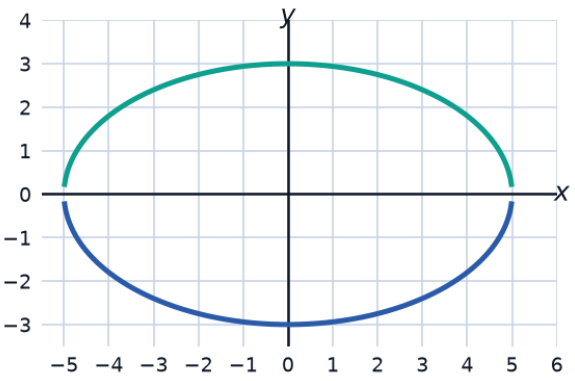
6 [Foundational] • Target: Parametric equations for lines and conics

Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.



7 [Foundational] • Target: Parametric equations for lines and conics

Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.

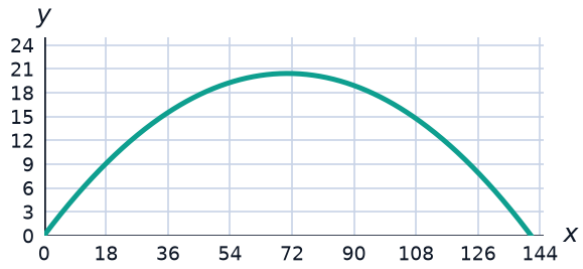


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Modeling projectile motion parametrically

A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)\,t$ and $y(t) = 40\sin(30^\circ)\,t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.

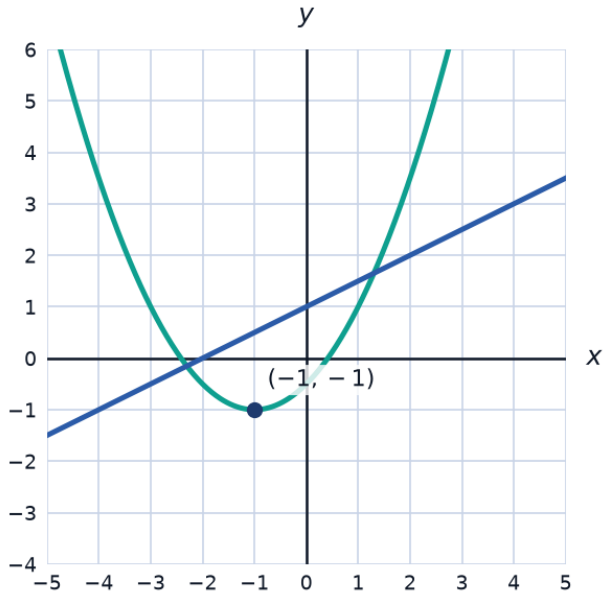


9 [Intermediate] • Target: Modeling projectile motion parametrically

For the same projectile, find the time at which the ball lands (returns to $y = 0$).

10 [Intermediate] • Target: Comparing parametric speed and direction

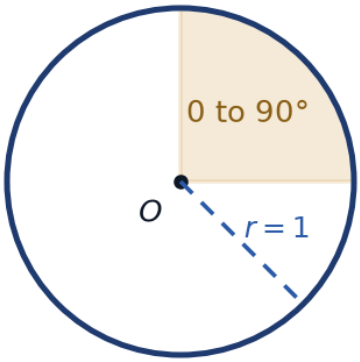
Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.



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11 **[Intermediate]** • **Target: Restricting the parameter's domain**

The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in [0, \frac{\pi}{2}]$.

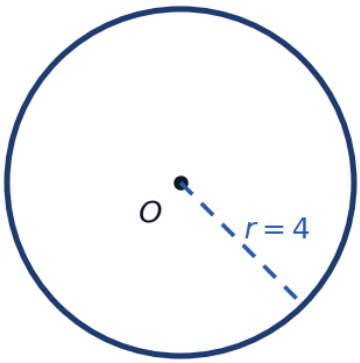


12 **[Intermediate]** • **Target: Restricting the parameter's domain**

Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.

13 **[Intermediate]** • **Target: Parametric equations for circular motion with a phase shift**

A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.



14 **[Intermediate]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = t - 6$ and $y(t) = -3t - 3$ to find the rectangular equation for y in terms of x.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Visualizing a parametric curve's components • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Eliminate the parameter from $x(t) = t + 9$ and $y(t) = 3t + 4$ to find the rectangular equation for y in terms of x.

16 [Challenging] • Target: Parametric equations for lines and conics • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Eliminate the parameter from $x(t) = t$ and $y(t) = 8t - 9$ to find the rectangular equation for y in terms of x.

17 [Challenging] • Target: Modeling projectile motion parametrically • Cross-Lesson: 4.01 Inverse & Composite Functions

Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -8t + 3$ to find the rectangular equation for y in terms of x.

18 [Challenging] • Target: Comparing parametric speed and direction • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -t - 5$ to find the rectangular equation for y in terms of x.

19 [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Given $x(t) = t^2 + 6$ and $y(t) = -2t + 8$, find the point on the curve at $t = 0$.

20 [Challenging] • Target: Parametric equations for circular motion with a phase shift • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Given $x(t) = t^2 + 2$ and $y(t) = -2t + 7$, find the point on the curve at $t = -1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating parametric equations

Given $x(t) = -3t - 2$ and $y(t) = -5t$, find the point on the curve at $t = -2$.

a. $(10, 4)$

b. $(4, 10)$

c. $(4, 11)$

d. $(5, 10)$

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22 [Foundational] • Target: Eliminating the parameter

Given $x(t) = -5t - 5$ and $y(t) = 2t - 3$, find the point on the curve at $t = 2$.

a. $(-15, 2)$

b. $(-14, 1)$

c. $(-15, 1)$

d. $(1, -15)$

23 [Foundational] • Target: Visualizing a parametric curve's components

Given $x(t) = -2t + 9$ and $y(t) = 3t + 5$, find the point on the curve at $t = 0$.

a. $(9, 5)$

b. $(9, 6)$

c. $(10, 5)$

d. $(5, 9)$

24 [Foundational] • Target: Parametric equations for lines and conics

Given $x(t) = -3t + 4$ and $y(t) = -3t + 4$, find the point on the curve at $t = -3$.

a. $(13, 13)$

b. $(11, 15)$

c. $(14, 13)$

d. $(13, 14)$

25 [Foundational] • Target: Modeling projectile motion parametrically

Given $x(t) = 4t - 7$ and $y(t) = t - 7$, find the point on the curve at $t = -5$.

a. $(-27, -12)$

b. $(-26, -12)$

c. $(-27, -11)$

d. $(-12, -27)$

26 [Foundational] • Target: Comparing parametric speed and direction

Given $x(t) = 6t + 9$ and $y(t) = 2t + 5$, find the point on the curve at $t = -2$.

a. $(-2, 1)$

b. $(-3, 1)$

c. $(-3, 2)$

d. $(1, -3)$

27 [Foundational] • Target: Restricting the parameter's domain

Given $x(t) = 4t + 4$ and $y(t) = t + 9$, find the point on the curve at $t = -6$.

a. $(-20, 4)$

b. $(-19, 3)$

c. $(3, -20)$

d. $(-20, 3)$

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28 [Foundational] • Target: Parametric equations for circular motion with a phase shift

Given $x(t) = 6t - 4$ and $y(t) = 5t - 3$, find the point on the curve at $t = 5$.

a. $(27, 22)$

b. $(22, 26)$

c. $(26, 22)$

d. $(26, 23)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Evaluating Parametric Equations and Graph Points

Eliminate the parameter from $x(t) = t - 9$ and $y(t) = -7t + 7$ to find the rectangular equation for y in terms of x.

a. $y = 7x - 56$

b. $y = -7x - 55$

c. $y = -7x + 7$

d. $y = -7x - 56$

30 [Intermediate] • Target: Eliminating the Parameter

Eliminate the parameter from $x(t) = t + 5$ and $y(t) = 7t - 2$ to find the rectangular equation for y in terms of x.

a. $y = 7x - 36$

b. $y = 7x - 37$

c. $y = 7x - 2$

d. $y = -7x - 37$

31 [Intermediate] • Target: Projectile Motion Modeling

Eliminate the parameter from $x(t) = t - 1$ and $y(t) = 8t - 5$ to find the rectangular equation for y in terms of x.

a. $y = 8x - 5$

b. $y = -8x + 3$

c. $y = 8x + 3$

d. $y = 8x + 4$

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32 [Intermediate] • Target: Direction of Motion and Speed

Eliminate the parameter from $x(t) = t - 5$ and $y(t) = t - 7$ to find the rectangular equation for y in terms of x.

a. $y = -x - 2$

b. $y = x - 7$

c. $y = x - 2$

d. $y = x - 1$

33 [Intermediate] • Target: Evaluating parametric equations

Eliminate the parameter from $x(t) = t + 6$ and $y(t) = t + 4$ to find the rectangular equation for y in terms of x.

a. $y = x - 2$

b. $y = -x - 2$

c. $y = x - 1$

d. $y = x + 4$

34 [Intermediate] • Target: Eliminating the parameter

Given $x(t) = t^2 - 9$ and $y(t) = 4t + 3$, find the point on the curve at $t = -1$.

a. $(-8, 1)$

b. $(-1, -8)$

c. $(-6, -1)$

d. $(-8, -1)$

35 [Intermediate] • Target: Visualizing a parametric curve's components

A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{11\pi}{6}$ (330°).

a. $(2\sqrt{3}, 2)$

b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(-2\sqrt{3}, -2)$

36 [Intermediate] • Target: Parametric equations for lines and conics

A curve is given by $x(t) = 8\cos t$, $y(t) = 8\sin t$. Find the exact point on the curve at $t = \frac{3\pi}{2}$ (270°).

a. $(0, -8)$

b. $(-8, 0)$

c. $(-4\sqrt{3}, -4)$

d. $(0, 8)$

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37 [Intermediate] • Target: Modeling projectile motion parametrically

A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{7\pi}{4}$ (315°).

a. $(2\sqrt{2}, 2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(-2\sqrt{2}, -2\sqrt{2})$

d. $(2\sqrt{2}, -2\sqrt{2})$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing parametric speed and direction • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A projectile is launched with initial speed 49 m/s at 60° above horizontal, so $y(t) = 49\sin(60^\circ)\,t - 4.9t^2$. Find $y(2)$, to one decimal place.

a. 65.3

b. 70.3

c. 67.3

d. 63.3

39 [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A projectile is launched with initial speed 60 m/s at 30° above horizontal, so $y(t) = 60\sin(30^\circ)\,t - 4.9t^2$. Find $y(1)$, to one decimal place.

a. 27.1

b. 25.1

c. 23.1

d. 30.1

40 [Challenging] • Target: Evaluating Parametric Equations and Graph Points • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A particle moves according to $x(t) = t^2 - 3$ and $y(t) = 2t + 1$. What is its position at $t = 3$?

a. $(6, 7)$

b. $(9, 7)$

c. $(6, 6)$

d. $(3, 7)$

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41 [Challenging] • Target: Evaluating Parametric Equations and Graph Points • Cross-Lesson: 1.03 Rates of Change & Concavity

Given $x(t) = 4\cos t$ and $y(t) = 4\sin t$ for $0 \le t \le 2\pi$, what curve is traced and in which direction?

a. A circle of radius 4 traced counterclockwise

b. A circle of radius 4 traced clockwise

c. An ellipse of semi-major axis 4

d. A line segment from $(-4, -4)$ to $(4, 4)$

42 [Challenging] • Target: Eliminating the Parameter • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Eliminate the parameter from $x = 3\cos\theta$ and $y = 5\sin\theta$.

a. $\frac{x^2}{9} + \frac{y^2}{25} = 1$

b. $\frac{x^2}{25} + \frac{y^2}{9} = 1$

c. $x^2 + y^2 = 34$

d. $9x^2 + 25y^2 = 1$

43 [Challenging] • Target: Projectile Motion Modeling • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A projectile is launched with initial speed $v_0 = 80$ ft/s at an angle of 30° from the horizontal from ground level. Which parametric equations model its path ($g = 32$ ft/s^2)?

a. $x(t) = 40\sqrt{3}t$ and $y(t) = 40t - 16t^2$

b. $x(t) = 40t$ and $y(t) = 40\sqrt{3}t - 16t^2$

c. $x(t) = 80t$ and $y(t) = 80t - 32t^2$

d. $x(t) = 40\sqrt{3}t$ and $y(t) = 40t - 32t^2$

44 [Challenging] • Target: Projectile Motion Modeling • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

What is the maximum height attained by a projectile modeled by $y(t) = 64t - 16t^2$?

a. 64 ft

b. 32 ft

c. 128 ft

d. 48 ft

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45 [Challenging] • Target: Direction of Motion and Speed • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

If a curve is given by $x(t) = t + 1$ and $y(t) = t^2$, what is the average rate of change $\frac{\Delta y}{\Delta x}$ from $t = 1$ to $t = 3$?

a. 4

b. 8

c. 2

d. 6

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